

LIFE PROGRAMME

LIFE Standard Action Project

Circular Textile Recovery Network for Mediterranean Cities LIFE-RECITEX

Field	Value
Call Identifier	LIFE-2026-SAP-ENV
Type of Action	LIFE Standard Action Project (SAP)
Duration	42 months
TRL Range	TRL 5 to TRL 8
Total Budget	EUR 3,000,000
EU Contribution	EUR 1,650,000

Abstract

LIFE-RECITEX demonstrates and scales a replicable municipal system for the separate collection, automated near-infrared (NIR) sorting, and fibre-to-fibre recycling of post-consumer textiles across three Mediterranean cities: Turin (IT), Valencia (ES), and Thessaloniki (GR). Working from a baseline separate collection rate of only 3.1 kg per inhabitant per year — against a theoretical recyclable potential of 9.5 kg per inhabitant per year — the project will deploy 240 smart IoT-enabled collection points, commission a NIR sorting facility with a validated throughput of 3,000 tonnes per year, and activate binding fibre-to-fibre recycling offtake agreements. Over 42 months, RECITEX will divert a minimum of 4,200 tonnes of textile waste from landfill and incineration, avoiding at least 6,300 tCO₂e of greenhouse gas emissions. IoT-driven dynamic collection routing will reduce collection vehicle kilometres by 25% relative to fixed-schedule baselines. A multilingual citizen engagement campaign will reach a minimum of 300,000 citizens across the three cities. A Municipal Textile Recovery Governance Framework and a comprehensive Replication Toolkit — both published open-access under CC-BY 4.0 — will be disseminated to at least 30 Mediterranean municipalities through dedicated policy workshops, with binding replication commitments sought from a minimum of five additional cities by project end, creating conditions for a further 15,000 tonnes per year of textile diversion post-project. RECITEX directly implements the EU Waste Framework Directive separate-collection obligation for textiles, contributes to the EU Strategy for Sustainable and Circular Textiles and the Circular Economy Action Plan, and provides CINEA and the European Commission with a validated reference model for monitoring WFD compliance across Member States.

Keywords: textile waste, separate collection, fibre-to-fibre recycling, near-infrared sorting, circular economy, Mediterranean cities, Waste Framework Directive, municipal waste management, zero pollution, replication

SECTION 1 - RELEVANCE AND OBJECTIVES

1.1 Objectives and Ambition

Post-consumer textiles constitute one of the fastest-growing waste streams in the European Union. According to the European Environment Agency, EU households generate approximately 12.6 kg of textile waste per person per year, of which less than 20% is separately collected. The remainder is predominantly landfilled or co-incinerated with mixed municipal solid waste, generating significant greenhouse gas emissions and permanently destroying material value. In the three target cities of LIFE-RECITEX — Turin (IT), Valencia (ES) and Thessaloniki (GR) — baseline characterisation studies confirm that separate collection rates for post-consumer textiles average only 3.1 kg per inhabitant per year, against a theoretical recyclable potential of 9.5 kg per inhabitant per year. This represents a combined annual diversion gap of approximately 8,400 tonnes of recyclable textile material across the three municipalities.

RECITEX pursues three primary objectives: (1) to demonstrate a replicable, technology-enabled municipal system for the separate collection, automated near-infrared (NIR) sorting, and fibre-to-fibre recycling of post-consumer textiles at meaningful urban scale; (2) to divert a minimum of 4,200 tonnes of textile waste from landfill and incineration over the project lifetime, corresponding to 50% of the identified diversion gap; and (3) to produce a validated governance and operational model enabling at least five additional Mediterranean municipalities to adopt equivalent systems within three years of project completion.

These objectives are explicitly SMART: they are time-bound to the 42-month project period; measurable through certified waste tonnage records and collection-point utilisation data; achievable on the basis of the consortium's demonstrated technical and institutional capacity; relevant to the EU Waste Framework Directive (WFD) separate-collection obligation for textiles; and directly attributable to project actions. Beyond the state of the art, RECITEX advances the field in three respects. First, whereas existing pilot initiatives have applied NIR sorting to mono-material plastic or paper fractions, RECITEX validates NIR classification across a minimum of eight mixed textile fibre-composition categories at commercial throughput — a capability not yet demonstrated at municipal scale in the Mediterranean context. Second, the integration of IoT fill-level sensing with dynamic logistics routing, applied specifically to textile collection, has not been validated across multi-city deployments in southern Europe. Third, the project generates the first harmonised, cross-national Municipal Textile Recovery Governance Framework specifically designed to support WFD transposition in Mediterranean administrative contexts.

1.2 Relation to the Work Programme

RECITEX is directly responsive to Call topic LIFE-2026-SAP-ENV-ENVIRONMENT, Priority Area 1 (Circular Economy and Waste), sub-topic 1.1(g): 'Separate collection, preparation for re-use and recycling of textiles, including mattresses and footwear.' The project addresses the core legislative implementation gap created by Article 22a of the revised EU Waste Framework Directive (WFD, Directive 2008/98/EC as amended by Directive 2018/851/EC), which obliges Member States to establish separate collection systems for textiles by 1 January 2025. Despite this obligation, implementation across Mediterranean cities remains fragmented and insufficiently supported by appropriate collection infrastructure and downstream sorting and recycling capacity. RECITEX bridges this implementation gap through concrete, cross-national demonstration action.

The project aligns directly with the Circular Economy Action Plan (CEAP, COM/2020/98) and the EU Strategy for Sustainable and Circular Textiles (COM/2022/141), both of which set targets for making textile products more durable, repairable, and recyclable, and for scaling up collection and recycling infrastructure. RECITEX also contributes to the Zero Pollution Action Plan by reducing the landfilling

and incineration of recyclable materials, and to the European Green Deal objective of decoupling economic activity from resource consumption.

EU added value is demonstrated at three levels. First, the project operates across three Member States at different stages of WFD transposition, generating comparative evidence and a harmonised operational model that no single national authority could develop in isolation. Second, the NIR-sorting technology benchmarks and fibre-to-fibre recycling partnership models will generate transferable lessons directly applicable to WFD implementation and enforcement across all 27 Member States. Third, the Replication Toolkit produced under WP5 is specifically designed to serve as a reference instrument for CINEA and the European Commission in monitoring compliance with the separate-collection mandate, ensuring that project outputs feed directly into EU-level governance.

1.3 Concept and Methodology

RECITEX adopts a systemic, demonstration-and-transfer methodology structured along three operational pillars: (i) smart collection infrastructure deployment; (ii) automated material processing and circular value-chain activation; and (iii) governance model codification and replication.

Pillar I — Smart Collection Infrastructure: Each of the three pilot cities will deploy a network of 80 smart collection points (240 total), equipped with IoT fill-level sensors and integrated citizen-reward interfaces. Sensor data will be aggregated on a cloud-based logistics platform to enable dynamic collection routing, reducing collection vehicle kilometres by an estimated 25% compared to fixed-schedule approaches. Siting analysis will be conducted using GIS-based accessibility modelling to ensure that a minimum of 80% of households in target districts are within 300 metres walking distance of a collection point. This evidence-based siting methodology goes beyond current practice, which typically relies on population density proxies rather than geospatial accessibility optimisation.

Pillar II — Automated Sorting and Recycling: Collected textiles will be transported to a centralised NIR sorting facility operated by EcoTessile Srl, capable of processing 3,000 tonnes per year and classifying textiles into a minimum of eight fibre-composition categories. Sorted streams will be channelled to contracted fibre-to-fibre recycling partners under pre-negotiated offtake agreements, ensuring that at least 60% of sorted material by weight enters closed-loop recycling rather than open-loop or energy-recovery pathways. Sorting performance will be validated by an independent laboratory against documented classification accuracy thresholds, and the methodology will draw on EN 15347 principles adapted by analogy to textile fractions.

Pillar III — Governance Model and Citizen Engagement: The Municipality of Valencia and the Hellenic Recycling Agency will co-develop and test a Municipal Textile Recovery Governance Framework, defining roles, responsibilities, procurement models, and performance benchmarks for public authorities. Citizen awareness campaigns, co-designed with local social enterprises and schools, will be deployed in all three cities, targeting a minimum reach of 300,000 citizens. The full methodology incorporates a Do No Significant Harm (DNSH) assessment conducted at project inception and applies EU green public procurement (GPP) criteria to all equipment and service contracts, ensuring environmental integrity throughout implementation.

1.4 Interdisciplinary Considerations

RECITEX explicitly integrates environmental engineering, behavioural science, urban planning, and public policy analysis. The consortium brings together an environmental SME with proven material-processing expertise (EcoTessile Srl), a large Mediterranean municipality with urban governance and citizen engagement capacity (Municipality of Valencia), and a national recycling authority with regulatory and cross-sectoral knowledge (Hellenic Recycling Agency). This disciplinary diversity is a deliberate design choice: textile waste diversion cannot be resolved by technology alone but requires

simultaneous change in citizen behaviour, urban logistics, regulatory frameworks, and secondary market structures. Interdisciplinary integration is operationalised through co-led work packages — in particular, WP2 and WP4, where technical infrastructure and citizen engagement are co-designed to ensure that collection infrastructure placement responds to behavioural accessibility insights rather than purely logistical criteria.

The project will apply open science principles throughout, publishing all NIR sorting algorithms, collection-point siting models, and governance templates under a Creative Commons CC-BY 4.0 licence, ensuring that outputs are freely accessible and adaptable by municipalities and researchers beyond the project consortium. Gender-disaggregated data will be collected in all citizen engagement activities to identify and address differential participation patterns by gender, in line with the European Commission's guidance on the gender dimension in research and innovation. Environmental footprint calculations for the project's own operations will follow the Product Environmental Footprint (PEF) methodology, ensuring that project activities themselves remain net-beneficial relative to a no-project baseline by Month 18 and throughout the remainder of the implementation period.

SECTION 2 - IMPACT AND EU ADDED VALUE

2.1 Pathways Towards Impact

RECITEX will generate quantified environmental impact through two direct pathways and one systemic replication pathway.

The first direct pathway is waste diversion: the deployment of 240 smart collection points across three cities, combined with intensive and measurable citizen engagement, is projected to increase the separate collection rate from a baseline of 3.1 kg per inhabitant per year to 7.5 kg per inhabitant per year by Month 36, diverting a cumulative minimum of 4,200 tonnes of post-consumer textiles from landfill and incineration over the project lifetime. Using IPCC Tier 1 emission factors for textile landfilling (0.45 tCO₂e per tonne) and incineration with mixed municipal solid waste (2.1 tCO₂e per tonne), this diversion corresponds to an estimated 6,300 tCO₂e avoided over the project period, a figure that will be independently verified at project close.

The second direct pathway is circular material output: NIR sorting will generate a minimum of 2,520 tonnes of fibre-to-fibre recyclable material (representing 60% of collected volumes) channelled to contracted recycling partners under binding offtake agreements, displacing an equivalent quantity of virgin fibre production and the associated water consumption, chemical inputs, and embedded carbon.

The systemic replication pathway operates through governance transfer: the Replication Toolkit and the Municipal Textile Recovery Governance Framework will be formally presented to a minimum of 30 Mediterranean municipalities through dedicated policy replication workshops, and binding replication commitment letters are targeted from at least five additional cities by project end. Conservative modelling based on the three pilot cities' diversion uplift indicates that adoption by five cities of comparable scale would generate a combined additional diversion potential of at least 15,000 tonnes per year post-project — a multiplier effect that substantially amplifies the direct environmental return on EU investment.

RECITEX is designed to avoid risk of substantial harm to other LIFE Programme objectives. No hazardous materials are introduced into any process stream; the NIR sorting process is solvent-free; IoT-enabled logistics optimisation actively reduces transport-related emissions; and all equipment procurement applies EU GPP energy-efficiency criteria. A Do No Significant Harm assessment will be completed and reported by Month 6.

2.2 Dissemination, Exploitation and Communication

Dissemination, communication, and exploitation activities are led by EcoTessile Srl with active contributions from all partners, and are structured around four target audiences: (1) citizens in the three pilot cities; (2) municipal administrations and waste management authorities across the Mediterranean and the wider EU; (3) the EU textile recycling industry and secondary fibre markets; and (4) EU policy makers and CINEA.

For citizens, the project will deploy a multilingual awareness campaign in Italian, Spanish, Greek, and English using social media channels, local press, structured school programmes, and digital interfaces integrated into the smart collection points themselves. The campaign targets a minimum of 300,000 citizens reached and a measurable increase in correct textile-sorting awareness of at least 40 percentage points in target districts, assessed through baseline and mid-term surveys.

For municipal administrations, three Policy Replication Workshops will be organised in Rome (Month 18), Valencia (Month 30), and Thessaloniki (Month 40), each targeting waste management officers

and elected officials from Mediterranean cities. Workshop proceedings and key findings will be published as open-access policy briefs on the project website and uploaded to the LIFE Project Database. The Replication Toolkit — comprising a Municipal Readiness Assessment Tool, standardised tender documents, and a financial viability model — will be released under a CC-BY 4.0 licence by Month 36 to maximise uptake and adaptation.

For the textile recycling industry, EcoTessile Srl will publish validated NIR sorting performance benchmarks and anonymised offtake agreement templates to lower barriers to market entry for operators seeking to develop fibre-to-fibre recycling capacity.

At EU level, project results will be presented at a minimum of two major EU-level events — such as the Circular Economy Stakeholder Conference and the RWM Exhibition — and submitted to at least three peer-reviewed journals to ensure scientific credibility and long-term archival accessibility.

Post-project sustainability is secured through three complementary mechanisms: (i) the Municipality of Valencia and the Hellenic Recycling Agency will formally embed the collection networks in their permanent municipal and national waste management plans by Month 40; (ii) EcoTessile Srl will continue NIR sorting facility operations on a commercial basis, with the offtake revenue model validated during the project period; and (iii) the governance framework will be submitted for formal adoption as a regional best-practice standard by relevant authorities in Italy, Spain, and Greece, ensuring institutional continuity beyond LIFE Programme funding.

2.3 Summary of Impact Indicators

Indicator	Target	Measurement
Tonnes of post-consumer textiles separately collected per year across three pilot cities	Increase from baseline 3.1 kg/inh/year to 7.5 kg/inh/year by Month 36; cumulative diversion $\geq 4,200$ t over project lifetime	Certified weight records from collection point emptying logs and transfer station weight bridges; reported semi-annually
CO ₂ -equivalent emissions avoided from diverted textile waste	Minimum 6,300 tCO ₂ e avoided cumulatively over 42 months	Calculated using IPCC Tier 1 emission factors for landfill and incineration; verified by independent environmental auditor at Month 42
Tonnes of recycled secondary fibre entering fibre-to-fibre recycling	Minimum 2,520 tonnes over project lifetime (60% of collected volume)	Offtake agreement delivery notes and recycler certification records; reported quarterly
Number of citizens reached by awareness campaign	Minimum 300,000 citizens across three cities	Campaign analytics (social media reach, event attendance registers, school programme records); reported semi-annually
Number of municipalities issuing formal replication commitment letters	Minimum 5 additional Mediterranean municipalities by Month 42	Signed commitment letters submitted to CINEA as project deliverable
Reduction in collection vehicle kilometres through IoT route optimisation	25% reduction vs. fixed-schedule baseline by Month 24	Fleet GPS data analysed and reported by logistics platform operator

SECTION 3 - IMPLEMENTATION, WORK PLAN AND RESOURCES

3.1 Work Plan and Work Packages

WP	Title	Lead	PM	Duration
WP1	Project Preparation and Baseline Assessment	EcoTessile Srl	18	M1-M6
WP2	Infrastructure Deployment and Operational Launch	Municipality of Valencia	42	M5-M24
WP3	Operations, Monitoring and Environmental Impact Measurement	Hellenic Recycling Agency	36	M13-M42
WP4	Citizen Engagement and Awareness	Municipality of Valencia	24	M6-M40
WP5	Replication, Dissemination and Communication	EcoTessile Srl	20	M12-M42
WP6	Project Management and Quality Assurance	EcoTessile Srl	24	M1-M42

WP1: Project Preparation and Baseline Assessment

Lead: EcoTessile Srl | **Person-months:** 18 | **Duration:** M1 - M6

WP1 covers all preparatory activities required before infrastructure deployment commences. This includes detailed baseline characterisation of textile waste generation and separate collection rates in each pilot city, GIS-based siting analysis for the 240 smart collection points, finalisation of the NIR sorting facility design, and procurement of IoT hardware and collection equipment in full compliance with EU green public procurement (GPP) criteria. Offtake agreements with fibre-to-fibre recycling operators are negotiated and signed in this WP to ensure downstream processing capacity is contractually secured before the first tonne is collected. Legal and administrative project setup, including the consortium agreement, data management plan, and DNSH assessment, is also completed within this WP. The outputs of WP1 are prerequisite gate conditions for the start of WP2 deployment activities.

T1.1: Baseline waste characterisation and GIS siting analysis - Conduct field campaigns and secondary data analysis to quantify textile waste generation and existing separate collection rates in Turin, Valencia, and Thessaloniki. Produce city-level GIS maps identifying optimal collection point locations to ensure that a minimum of 80% of households in target districts are served within 300 metres walking distance. Baseline data will serve as the reference values for all project environmental indicators.

T1.2: Infrastructure procurement and supplier selection - Issue open tenders for smart collection points, NIR sorting equipment, and the IoT logistics platform in full compliance with EU GPP criteria for energy efficiency and material durability. Evaluate bids, finalise supplier contracts, and establish delivery and installation schedules aligned with the WP2 phased deployment plan.

T1.3: Offtake agreement negotiation - Negotiate and sign binding offtake agreements with a minimum of three certified fibre-to-fibre recycling operators operating in different market segments (apparel, industrial, insulation) to guarantee diversified downstream processing capacity for at least 2,520 tonnes of sorted fibre over the project lifetime. Agreements will include minimum volume commitments, quality specifications, and pricing mechanisms.

WP2: Infrastructure Deployment and Operational Launch

Lead: Municipality of Valencia | **Person-months:** 42 | **Duration:** M5 - M24

WP2 manages the physical deployment of all 240 smart collection points across the three pilot cities, commissioning of the NIR sorting line at EcoTessile Srl's facility, integration of IoT fill-level sensors with the cloud-based logistics platform, and the full operational launch of the collection–sorting–recycling chain. Deployment is structured in two tranches of 120 units each (Months 5–12 and Months 13–24), with 40 units per city per tranche. This phased approach is deliberate: operational

data and citizen feedback from Tranche 1 inform siting and configuration adjustments for Tranche 2, reducing deployment risk and improving overall network performance. The Municipality of Valencia leads this WP by virtue of its urban project management capacity and its role as host city for the largest collection point network. Turin and Thessaloniki participate as co-implementers with equivalent deployment responsibilities in their respective city territories.

T2.1: Smart collection point installation — Tranche 1 - Install and commission 120 smart collection points (40 per city) at GIS-optimised locations identified in T1.1. Integrate IoT fill-level sensors with the cloud logistics platform and conduct end-to-end operational testing. Deliver staff training for municipal waste operators in all three cities. Document installation lessons learned to inform Tranche 2 configuration.

T2.2: Smart collection point installation — Tranche 2 - Install the remaining 120 collection points, incorporating siting and configuration lessons from Tranche 1. Achieve full network operational status — 240 points active and integrated across three cities — by Month 24, in alignment with Milestone MS5.

T2.3: NIR sorting line commissioning and validation - Commission the NIR sorting facility, conduct structured performance validation trials, and achieve certified classification accuracy of at least 90% across all eight target fibre categories. Independent laboratory validation will confirm throughput capacity of 3,000 tonnes per year and provide the technical basis for Deliverable D2.2. Contractual performance guarantees from the equipment supplier will be invoked if validation targets are not met within the commissioning period.

WP3: Operations, Monitoring and Environmental Impact Measurement

Lead: Hellenic Recycling Agency | Person-months: 36 | Duration: M13 - M42

WP3 covers the full operational running of the collection–sorting–recycling system from Month 13 through to project closure, together with continuous monitoring of all LIFE project indicators, quantified environmental impact assessment against the established baseline, and preparation of semi-annual monitoring reports. The Hellenic Recycling Agency leads this WP on the basis of its institutional mandate in regulatory monitoring and environmental compliance verification. An independent external auditor commissioned under this WP verifies impact data at Month 30 and at Month 42, providing CINEA with third-party assurance of reported environmental outcomes. WP3 incorporates an adaptive management cycle: monitoring data are reviewed quarterly by the Project Management Unit, enabling timely corrective action if collection rates or sorting performance deviate from trajectory.

T3.1: Collection and sorting operations - Manage operational running of the 240-point collection network and the NIR sorting facility, including dynamic vehicle route optimisation using IoT platform data, quality control of incoming textile streams, and timely delivery of sorted fibre fractions to offtake recycling partners in accordance with contractual volume and quality specifications.

T3.2: Environmental impact monitoring and LIFE KPI reporting - Collect, verify, and report all quantitative environmental key performance indicators semi-annually, including diverted tonnages certified by weight bridge records, CO₂-equivalent emissions avoided calculated using IPCC Tier 1 factors, secondary fibre output volumes confirmed by recycler delivery notes, and collection vehicle kilometre reductions validated by fleet GPS data. Commission and manage independent third-party verification at Month 30 and Month 42.

T3.3: Adaptive management and corrective action - Analyse monitoring data on a quarterly basis to identify underperforming collection points, sorting categories, or logistics routes. Implement corrective actions — including collection point relocation, citizen engagement intensification, or route reconfiguration — within defined response timescales. Document all corrective actions and outcomes in the project lessons-learned register, which will feed directly into the Replication Toolkit developed in WP5.

WP4: Citizen Engagement and Awareness

Lead: Municipality of Valencia | Person-months: 24 | Duration: M6 - M40

WP4 designs and implements multilingual citizen awareness and behavioural change campaigns across all three pilot cities, with the overarching target of reaching a minimum of 300,000 citizens and increasing correct textile-sorting awareness by at least 40 percentage points in target districts. Activities span social media campaigns, school textile literacy programmes, community events co-located at collection points, and a citizen reward scheme integrated with the IoT collection point digital interface. Campaign design draws on behavioural science insights to move beyond information provision towards active behaviour change, and co-design with local social enterprises ensures cultural and linguistic relevance in each city context. Gender-disaggregated participation data are collected throughout to ensure the campaign reaches all demographic groups equitably and to

generate evidence on differential engagement patterns relevant to future campaigns elsewhere.

T4.1: Campaign design and baseline awareness survey - Co-design multilingual campaign materials with local social enterprises and school representatives in each city, ensuring cultural and contextual relevance. Conduct a pre-campaign baseline awareness survey in target districts to establish measurable starting points for awareness and participation indicators. Survey methodology will enable gender-disaggregated analysis.

T4.2: Campaign implementation and school programme rollout - Deploy social media, local press, and community event campaigns in Italian, Spanish, Greek, and English. Implement structured textile literacy modules in a minimum of 30 schools across the three cities. Operate the citizen reward scheme through the IoT collection point interface, tracking participation rates and reward redemption to measure behavioural uptake.

T4.3: Impact evaluation and mid-term campaign adjustment - Conduct a mid-term awareness and participation survey at Month 24, analysing results by city, district, and gender. Adjust campaign messaging, channels, and incentive structures based on findings. Publish mid-term evaluation results, including gender-disaggregated data, as a public deliverable to contribute to the wider evidence base on textile waste behaviour change.

WP5: Replication, Dissemination and Communication

Lead: EcoTessile Srl | Person-months: 20 | Duration: M12 - M42

WP5 translates project results into transferable, publicly accessible outputs that maximise uptake beyond the three pilot cities and beyond the LIFE Programme funding period. The WP produces the Municipal Textile Recovery Governance Framework (Month 30) and the comprehensive Replication Toolkit (Month 36), organises three Policy Replication Workshops targeting Mediterranean municipal authorities, manages all external project communications, and secures binding replication commitment letters from at least five additional municipalities by project close. All outputs are published open-access under a Creative Commons CC-BY 4.0 licence. The WP is led by EcoTessile Srl, which combines technical credibility with an established European network in the textile recycling sector, while the Municipality of Valencia and the Hellenic Recycling Agency contribute peer-to-peer municipal and institutional outreach capacity essential for engaging target replication audiences.

T5.1: Replication Toolkit development - Develop the complete Replication Toolkit comprising: a Municipal Readiness Assessment Tool enabling self-assessment of collection infrastructure, institutional, and financial readiness; standardised EU GPP-compliant procurement template documents for collection points and NIR sorting equipment; and a financial viability model with sensitivity analysis covering different city sizes and co-funding scenarios. Release the final Toolkit under CC-BY 4.0 on the project website and the LIFE Project Database by Month 36.

T5.2: Policy Replication Workshops - Organise three structured Policy Replication Workshops: Rome (Month 18), Valencia (Month 30), and Thessaloniki (Month 40), each targeting waste management officers and decision-makers from Mediterranean municipalities. Actively follow up with workshop participants to secure signed replication commitment letters, with a minimum target of five letters received by Month 40 and submitted to CINEA as a formal project deliverable.

T5.3: Project website, publications and EU-level dissemination - Maintain an up-to-date project website with public access to all deliverables and monitoring data. Publish semi-annual project newsletters. Submit project results to a minimum of three peer-reviewed scientific journals to ensure academic credibility and open-archive accessibility. Present project results at a minimum of two major EU-level events such as the Circular Economy Stakeholder Conference and RWM Exhibition.

WP6: Project Management and Quality Assurance

Lead: EcoTessile Srl | Person-months: 24 | Duration: M1 - M42

WP6 provides overall coordination, financial management, quality assurance, risk management, and CINEA reporting throughout the full 42-month project period. EcoTessile Srl acts as Coordinating Beneficiary and hosts a dedicated Project Management Unit (PMU) staffed by a Project Manager and a Financial and Administrative Officer from Month 1. A Steering Committee comprising senior representatives of all three beneficiaries meets quarterly to review progress against the work plan, approve corrective actions, and take partnership-level decisions. The risk register is maintained and reviewed monthly by the PMU. All deliverables undergo internal quality review against defined acceptance criteria before external submission. Semi-annual progress reports and a comprehensive final report are submitted to CINEA in accordance with the grant agreement schedule.

T6.1: Project coordination and financial management - Coordinate day-to-day work plan execution across all WPs and partners; manage financial flows, inter-partner transfer procedures, and procurement compliance; maintain and update the risk register on a monthly basis; and convene quarterly Steering Committee meetings with documented minutes and action tracking.

T6.2: CINEA reporting and quality assurance - Prepare and submit all mandatory CINEA progress, financial, and final reports within grant agreement deadlines. Conduct internal quality assurance review of all project deliverables prior to submission, verifying compliance with technical specifications, open science requirements, and LIFE Programme communication guidelines.

3.2 Consortium Composition

Organisation	Country	Type	Role	Description
EcoTessile Srl	Italy	Private SME — environmental services and material recovery	Coordinator	Coordinating Beneficiary; leads overall project management, NIR sorting facility operations, WP1 preparatory actions, WP5 replication and dissemination, and WP6 management. Brings 12 years of expertise in textile waste processing and circular economy consultancy.
Municipality of Valencia	Spain	Public authority — local government	Partner	Associated Beneficiary; leads WP2 infrastructure deployment and WP4 citizen engagement. Contributes urban governance expertise, access to public land for collection point siting, and a citizen base of 800,000 for awareness campaigns.
Hellenic Recycling Agency	Greece	Public body — national recycling authority	Partner	Associated Beneficiary; leads WP3 operations and environmental monitoring. Contributes regulatory expertise, established relationships with the Greek recycling sector, and national-level dissemination capacity for replication in Greece.

The RECITEX consortium spans three Mediterranean EU Member States — Italy, Spain, and Greece — that are at demonstrably different stages of WFD Article 22a transposition and separate textile collection implementation. This geographic and regulatory diversity is a deliberate and essential design feature: it maximises the comparative learning value of the demonstration, produces a governance model tested under genuinely heterogeneous institutional and operational conditions, and ensures that project outputs carry credibility and direct applicability across the full range of Mediterranean administrative contexts.

Each partner contributes a distinct and non-substitutable competence. EcoTessile Srl provides the technical and commercial core of the project: 12 years of specialised expertise in textile waste

processing, NIR sorting technology operation, and circular economy consultancy, combined with the commercial continuity commitment that ensures the NIR sorting facility operates beyond the project period. The Municipality of Valencia contributes real-world urban governance testing at scale, with a citizen base of 800,000, established relationships with local social enterprises for co-designed engagement, and direct access to public land and permitting processes essential for rapid infrastructure deployment. The Hellenic Recycling Agency adds the regulatory and institutional depth that is indispensable for producing a governance framework credible to public authorities: its mandate encompasses national-level compliance monitoring, its established relationships with the Greek recycling sector underpin the Southern European replication pathway, and its national dissemination capacity amplifies project reach beyond the Thessaloniki pilot.

No bilateral or single-country partnership could replicate this transnational comparative dimension. The consortium is appropriately sized for a LIFE SAP — three committed beneficiaries with clearly delineated roles, no unnecessary overlap, and a governance structure that enables efficient decision-making — while providing the cross-national reach required to demonstrate EU added value and to generate a Replication Toolkit with genuine pan-Mediterranean applicability.

3.3 Milestones

ID	Title	WP	Due	Verification
MS1	Baseline characterisation and siting analysis complete	WP1	M4	Validated baseline report accepted by Steering Committee and submitted to CINEA
MS2	All procurement contracts signed and offtake agreements in force	WP1	M6	Signed contracts and offtake agreements submitted as annexes to first progress report
MS3	Tranche 1: 120 smart collection points operational across three cities	WP2	M12	Commissioning certificates and IoT platform operational dashboard screenshots
MS4	NIR sorting facility validated and at full throughput capacity	WP2	M14	Independent laboratory validation report confirming $\geq 90\%$ classification accuracy and 3,000 t/year capacity
MS5	Full collection network (240 points) operational and 1,000 tonnes diverted	WP3	M24	Weight bridge certified tonnage records; logistics platform cumulative collection data
MS6	Mid-term awareness survey completed; campaign adjusted	WP4	M25	Survey report with gender-disaggregated data submitted to CINEA
MS7	Replication Toolkit published open-access	WP5	M36	Toolkit URL live on project website; CC-BY 4.0 licence confirmed
MS8	Minimum 5 replication commitment letters received	WP5	M40	Signed commitment letters submitted to CINEA as formal project deliverable

MS9	Final environmental impact verified by independent auditor	WP3	M42	Independent audit report confirming ≥4,200 t diverted and ≥6,300 tCO ₂ e avoided
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3.4 Deliverables

ID	Title	WP	Due	Type	Dissemination
D1.1	Baseline Characterisation and GIS Siting Report	WP1	M4	Report	Public
D1.2	Procurement and Offtake Agreement Documentation Package	WP1	M6	Administrative document	Restricted
D2.1	Tranche 1 Commissioning Report — 120 Collection Points	WP2	M12	Technical report	Public
D2.2	NIR Sorting Line Validation Report	WP2	M14	Technical report	Public
D2.3	Full Network Commissioning Report — 240 Collection Points	WP2	M24	Technical report	Public
D3.1	Semi-Annual Environmental Monitoring Reports (x6)	WP3	M42	Monitoring report	Public
D3.2	Independent Environmental Impact Audit Report	WP3	M42	Audit report	Public
D4.1	Baseline and Mid-Term Citizen Awareness Survey Reports	WP4	M25	Report	Public
D4.2	Citizen Engagement Campaign Package (multilingual)	WP4	M18	Communication material	Public
D5.1	Municipal Textile Recovery Governance Framework	WP5	M30	Policy document	Public
D5.2	Replication Toolkit (Municipal Readiness Assessment Tool, procurement templates, financial model)	WP5	M36	Tool/Toolkit	Public
D5.3	Policy Replication Workshop Proceedings (x3)	WP5	M40	Report	Public
D6.1	Semi-Annual	WP6	M42	Administrative	Restricted

	Progress and Financial Reports to CINEA (x6)			report	
D6.2	Final Project Report	WP6	M42	Report	Public

3.5 Resources and Budget Summary

Cost Category	Amount (EUR)
Personnel Costs	1350000
Subcontracting	600000
Travel & Subsistence	90000
Equipment	650000
Other Direct Costs	160000
Indirect Costs (25%)	150000
TOTAL	3000000

3.6 Risk Management

ID	Risk	Likelihood	Impact	Mitigation
R1	Citizen participation rates fall below projections, resulting in collected tonnages below target	Medium	High	Phased campaign deployment with mid-term adjustment trigger at Month 24; citizen reward scheme to incentivise participation; contingency budget of EUR 30,000 allocated for additional campaign activities if M18 collection rates are below 50% of target.
R2	NIR sorting technology throughput or classification accuracy falls below validated specifications	Low	High	Independent laboratory validation at commissioning (D2.2) with contractual performance guarantees from equipment supplier; contingency manual pre-sorting protocol defined in operations manual; supplier obligated to remediate within 60 days of any identified underperformance.
R3	Secondary market demand for recycled fibre weakens, reducing offtake volumes and economic viability of recycling pathway	Medium	Medium	Minimum three diversified offtake agreements with recyclers in different market segments (apparel, industrial rags, insulation); project budget does not depend on offtake revenues; governance framework explicitly

				addresses market development strategies.
R4	Permitting or land-use delays slow collection point installation, pushing deployment milestones beyond schedule	Medium	Medium	Municipality of Valencia and Hellenic Recycling Agency engage permitting authorities in Month 1 as part of WP1 preparatory actions; buffer of two months built into Tranche 1 timeline; Tranche 2 timeline adjusted based on Tranche 1 experience.
R5	Consortium partner withdrawal or capacity reduction due to organisational change	Low	High	Consortium agreement includes step-in clauses enabling EcoTessile Srl to assume partner tasks with subcontractor support; project insurance covers partnership risk; critical knowledge is documented in shared project management platform accessible to all partners from Month 1.